

1 v0.8 치수·검사 조립 도면 세트

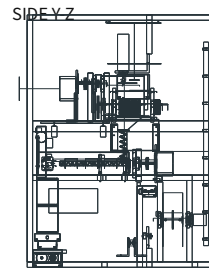
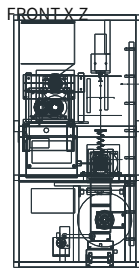
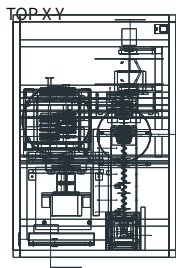
물리 검증·안전 인증·통전 승인이 아니다. E-stop, lid/service interlock, branch fuse, 독립 thermal fuse를 정상 firmware와 독립 구현하고 exact received component 정격·배선·보호소자를 실측 확인하기 전 통전하지 않는다.

Revision: final-design-fabrication-closure-v0.8 · 상태: DIGITAL_DOCUMENT / PHYSICAL_NOT_RUN / USER_APPROVAL_REQUIRED

2 GA-001 — general arrangement

GA-001 general arrangement

v0.8 · mm · 제3각법 · source: GGM integrated FreeCAD assembly · bodies 254



Overall X 470.00 · Y 729.00 · Z 930.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선

Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

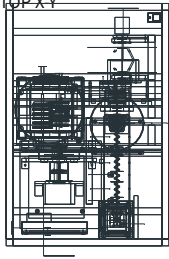
재료/구성: mixed assembly; see BOM.csv 핵심 치수/공차: 470 × 700 × 930 mm envelope; service aisle ≥ 600 mm 단위: mm · 제3각법 · NTS · source 2b085b5d04dda132909bd0e462a596893ad64912

3 ASM-001 — full assembly

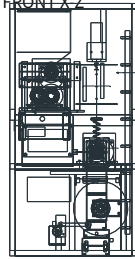
ASM-001 full assembly

v0.8 · mm · 제3각법 · source: GGM integrated FreeCAD assembly · bodies 254

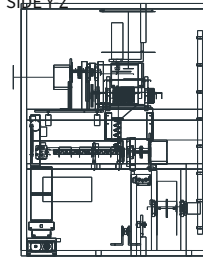
TOP X-Y



FRONT X-Z



SIDE Y-Z



Overall X 470.00 · Y 729.00 · Z 930.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

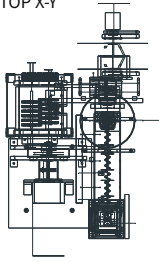
재료/구성: mixed assembly; see BOM.csv **핵심 치수/공차:** high-load path cutter/screw → metal bearing plate → profile → table **단위:** mm · 제3각법 · NTS · source 2b085b5d04dda132909bd0e462a596893ad64912

4 ASM-002 — module arrangement

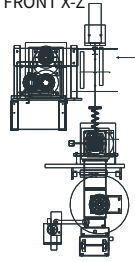
ASM-002 module arrangement

v0.8 · mm · 제3각법 · source: GGM integrated FreeCAD assembly · bodies 194

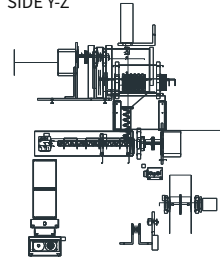
TOP X-Y



FRONT X-Z



SIDE Y-Z



Overall X 432.40 · Y 715.00 · Z 859.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

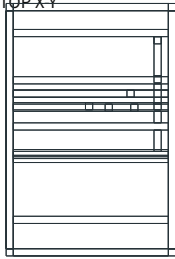
재료/구성: mixed assembly; see BOM.csv **핵심 치수/공차:** module datum transfer ≤ 0.50 mm; service modules removable without hot-path disturbance **단위:** mm · 제3각법 · NTS · source 2b085b5d04dda132909bd0e462a596893ad64912

5 FR-001 — frame

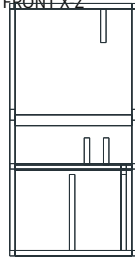
FR-001 frame

v0.8 · mm · 제3각법 · source: GGM integrated FreeCAD assembly · bodies 38

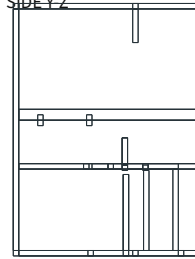
TOP X-Y



FRONT X-Z



SIDE Y-Z



Overall X 470.00 · Y 700.00 · Z 930.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

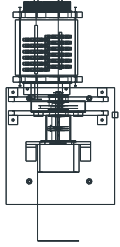
재료/구성: 2020/2040 aluminum profile **핵심 치수/공차:** base 470 × 700 mm; anchor M8 × 4; rail squareness ≤ 0.50/700
단위: mm · 제3각법 · NTS · source 2b085b5d04dda132909bd0e462a596893ad64912

6 SH-001 — shredder assembly

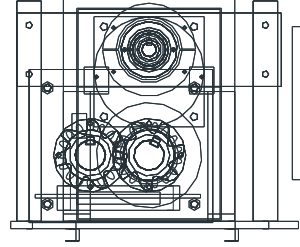
SH-001 shredder assembly

v0.8 · mm · 제3각법 · source: GGM integrated FreeCAD assembly · bodies 74

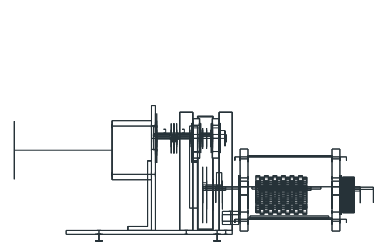
TOP X-Y



FRONT X-Z



SIDE Y-Z



Overall X 242.00 · Y 547.00 · Z 206.17 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

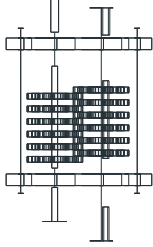
재료/구성: steel cutter module **핵심 치수/공차:** shaft centres 48.00 ± 0.03 mm; rotating-to-static clearance ≥ 1.90 mm; CUT-08 retainer $M4 \times 12 = 3$ N·m; CUT-03 pair tied by $4 \times$ CUT-09 OD10/ID6.6/L128 matched steel sleeves and $M6 \times 170$ class10.9 at 7 N·m; plates mount through G1J-10 feet **단위:** mm · 제3각법 · NTS · source 2b085b5d04dda132909bd0e462a596893ad64912

7 SH-002 — cutter stack

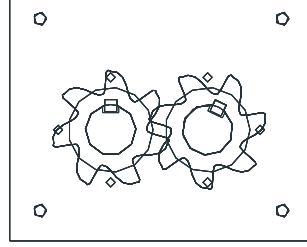
SH-002 cutter stack

v0.8 · mm · 제3각법 · source: GGM integrated FreeCAD assembly · bodies 20

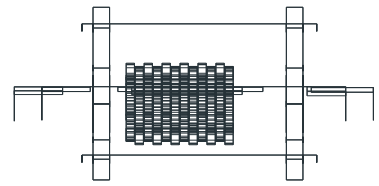
TOP X-Y



FRONT X-Z



SIDE Y-Z



Overall X 150.00 · Y 260.00 · Z 125.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

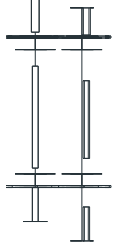
재료/구성: D2 cutters / numbered match-ground steel spacers **핵심 치수/공차:** CUT-01 t6 and CUT-02 nominal t7; do not interchange spacers; every one of 11 axial gaps 0.25–0.50 mm over one full hand rotation by metal shim **단위:** mm · 제3각법 · NTS · source 2b085b5d04dda132909bd0e462a596893ad64912

8 SH-003 — shaft and bearing assembly

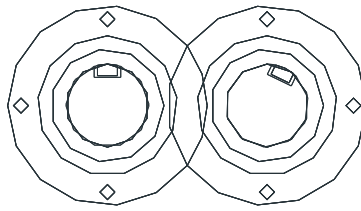
SH-003 shaft bearing

v0.8 · mm · 제3각법 · source: GGM integrated FreeCAD assembly · bodies 8

TOP X-Y



FRONT X-Z



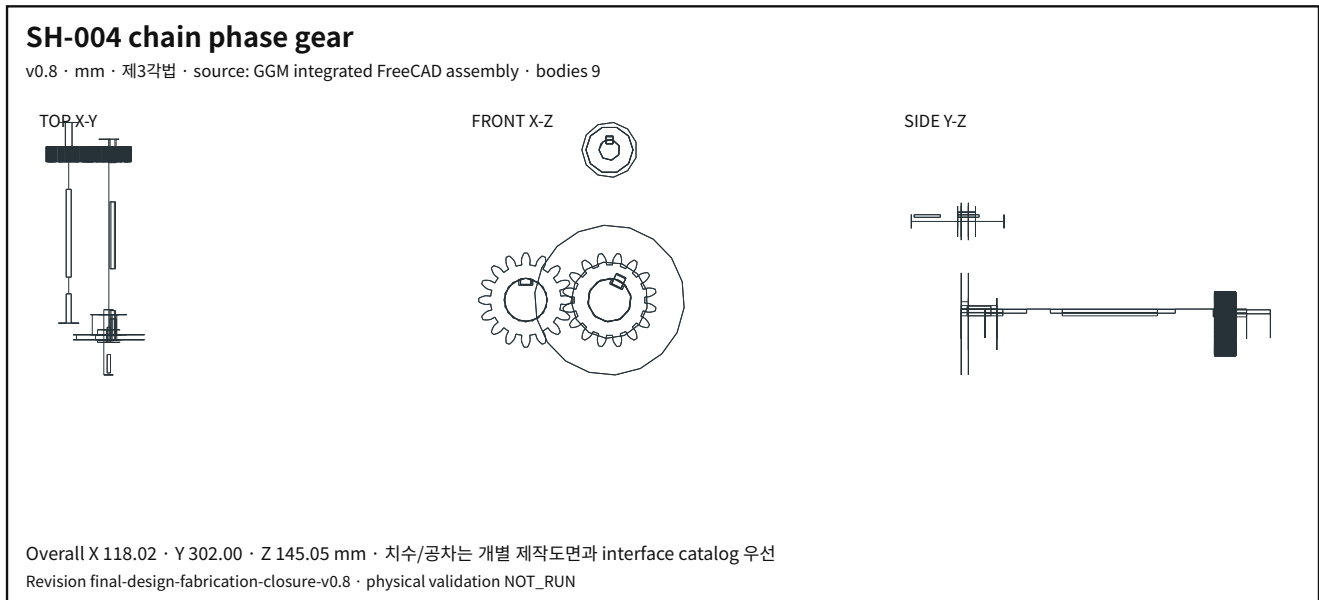
SIDE Y-Z



Overall X 108.00 · Y 260.00 · Z 60.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

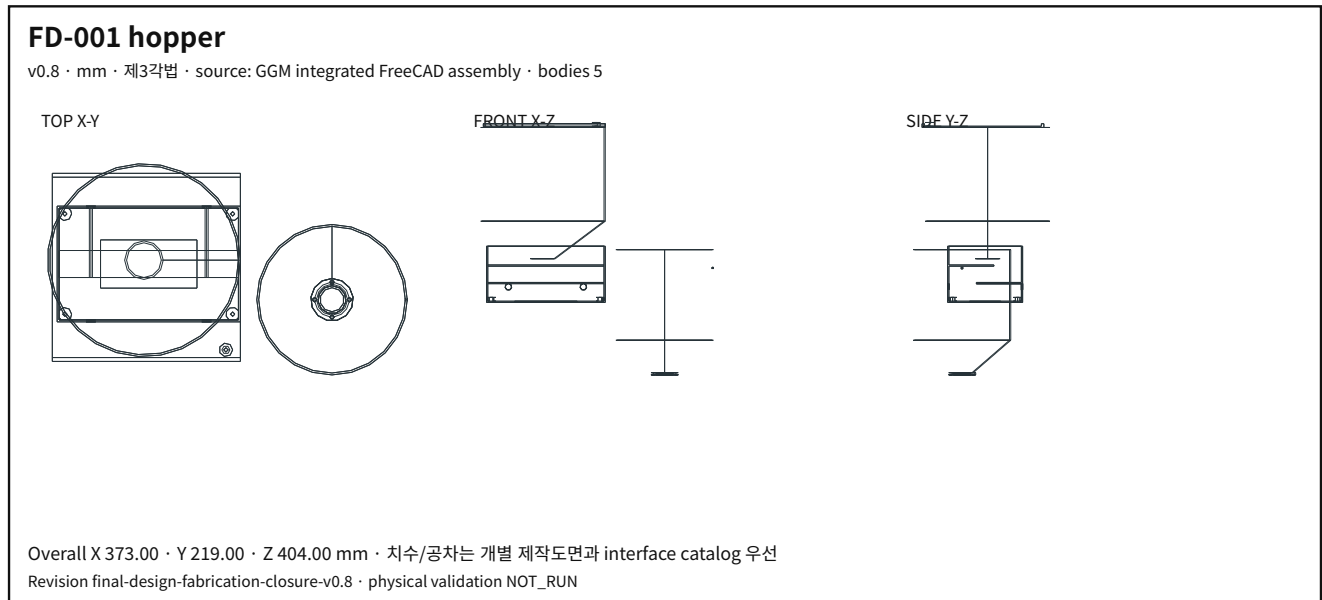
재료/구성: S45C QT shafts / SKF 61905-2RS1 핵심 치수/공차: Ø25 h6 seats; CUT-05R key clock $25.714 \pm 0.02^\circ$; shaft TIR ≤ 0.05 mm; centre parallelism $\leq 0.10/150$ 단위: mm · 제3각법 · NTS · source 2b085b5d04dda132909bd0e462a596893ad64912

9 SH-004 — chain and phase gear



재료/구성: direct-keyed steel 35 sprockets / S45C phase gears / 35 chain **핵심 치수/공차:** GGM_SH_12T direct-keyed to 12 h6 jackshaft with 4x4 key; GGM_SH_30T direct-keyed to CUT-05R 25 h6 with 6x6 key; keys carry torque; received maker retention hardware is axial-only and torque remains HOLD until receipt; sprocket radial TIR ≤ 0.10 mm; total axial shift+U95 ≤ 0.20 mm; chain alignment $\leq 0.20/150$ mm; midspan slack 2–3%; DRV-02 superseded; phase stack $M4 \times 4 = 3$ N·m **단위:** mm · 제3각법 · NTS · source 2b085b5d04dda132909bd0e462a596893ad64912

10 FD-001 — hopper



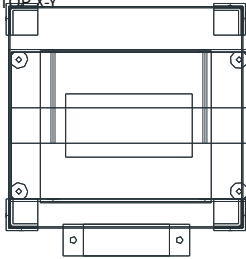
재료/구성: 5052-H32 hopper **핵심 치수/공차:** feed opening 150×150 mm; all reachable edges $R/C \geq 0.5$ mm **단위:** mm · 제3각법 · NTS · source 2b085b5d04dda132909bd0e462a596893ad64912

11 FD-002 — recirculation/screen

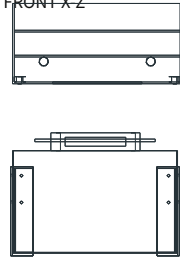
FD-002 recirculation screen

v0.8 · mm · 제3각법 · source: GGM integrated FreeCAD assembly · bodies 8

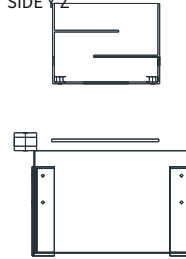
TOP X-Y



FRONT X-Z



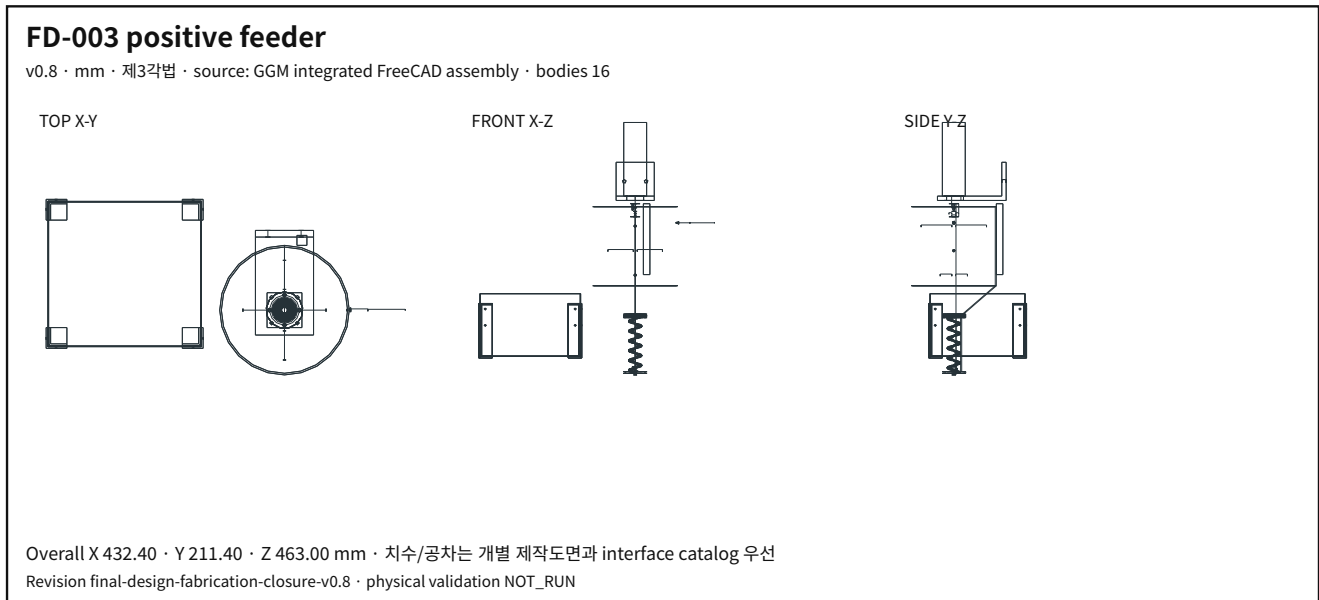
SIDE Y-Z



Overall X 192.00 · Y 199.40 · Z 283.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

재료/구성: 304 screen / sheet chute **핵심 치수/공차:** screen aperture Ø5 on 9 pitch; cutter/static clearance ≥ 1.90 mm 단
위: mm · 제3각법 · NTS · source 2b085b5d04dda132909bd0e462a596893ad64912

12 FD-003 — positive feeder



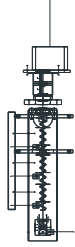
재료/구성: 304 auger/housing/common agitator shaft **핵심 치수/공차:** auger OD24.60; housing ID25.00 +0.05/0; radial clearance 0.20–0.25 mm; shaft/bore diametral clearance 0.200–0.322; SYS-15 Ø3x12 spring pin; pitch 18 **단위:** mm · 제3각법 · NTS · source 2b085b5d04dda132909bd0e462a596893ad64912

13 EX-001 — extruder assembly

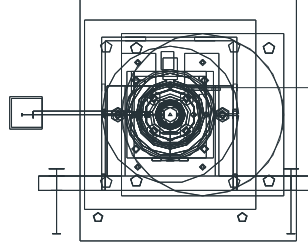
EX-001 extruder assembly

v0.8 · mm · 제3각법 · source: GGM integrated FreeCAD assembly · bodies 53

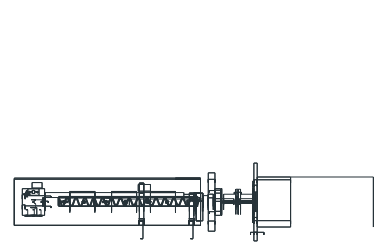
TOP-X-Y



FRONT X-Z



SIDE Y-Z



Overall X 168.40 · Y 646.00 · Z 140.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

재료/구성: SCM440 screw/barrel / steel supports **핵심 치수/공차:** integral barrel shoulder + rear retainer; 2× M4×25 class8.8 at 2.9 N·m; cold endplay 0.12–0.28 mm; front guide axial travel ≥ 1.50 mm; mount M5×4 = 2.5 N·m **단위:** mm
· 제3각법 · NTS · source 2b085b5d04dda132909bd0e462a596893ad64912

14 EX-002 — screw/barrel/die

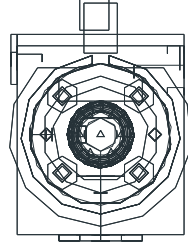
EX-002 screw barrel die

v0.8 · mm · 제3각법 · source: GGM integrated FreeCAD assembly · bodies 15

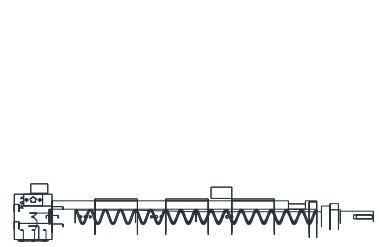
TOP X-Y



FRONT X-Z



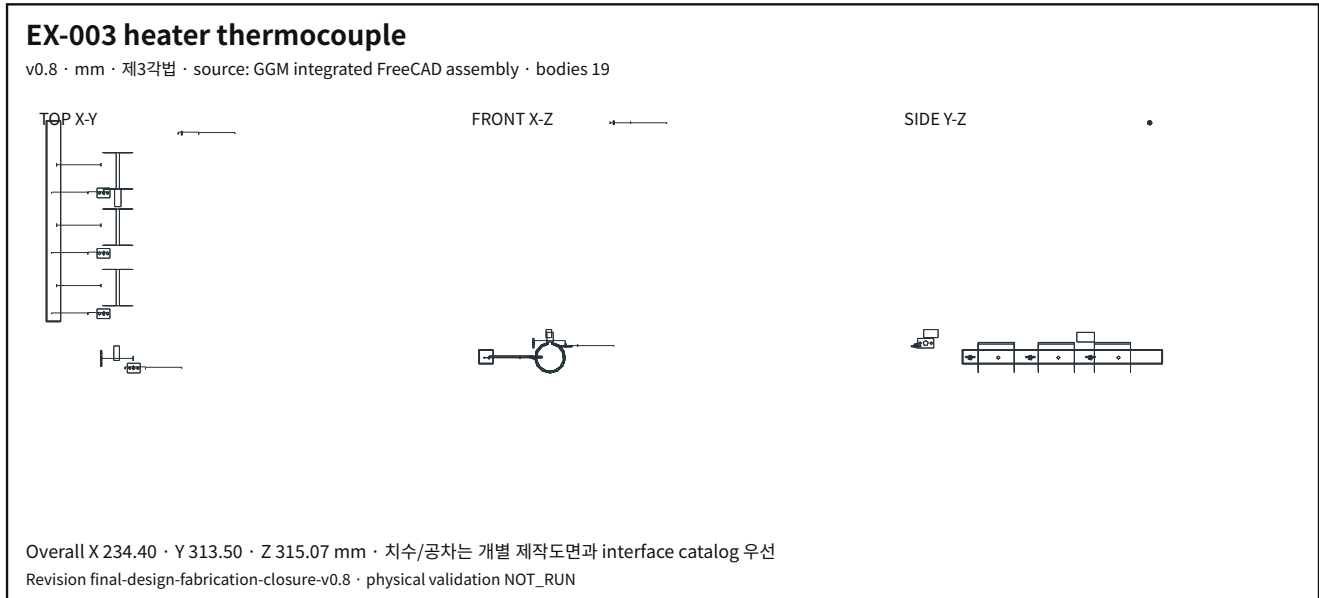
SIDE Y-Z



Overall X 44.00 · Y 380.50 · Z 60.50 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

재료/구성: nitrided SCM440 / NSK 51102 / 17-4PH die insert **핵심 치수/공차:** cold screw/barrel diametral clearance 0.28–0.32 mm; 51102 pocket clearance 0.30–0.35 mm; shaft abutment ≥ 23 mm; housing abutment ≤ 20 mm; shim-set loaded-direction endplay 0.05–0.15 mm; coaxiality ≤ 0.05 mm; thrust $M6 \times 8 = 9$ N·m; die SYS-04 $M4 \times 45 \times 4$ stock screws cut/deburred to 42.5 ± 0.1 give engagement 6.82–7.40/thread-bottom clearance 0.60–1.18; digital torque 1.50 N·m, physical receipt/leak/first thermal cycle NOT_RUN; retainer $M4 \times 2 = 1.2$ N·m **단위:** mm · 제3각법 · NTS · source 2b085b5d04dda132909bd0e462a596893ad64912

15 EX-003 — heater/thermocouple layout



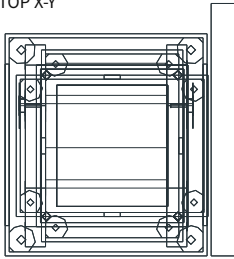
재료/구성: mica/NiCr heater and thermocouple candidates **핵심 치수/공차:** Band free-state ID34.10–34.20 and usable closure ≥ 1.00 leave worst-case closure reserve0.277 mm; after cold tightening, 0.05 mm feeler penetration ≤ 5 mm at 8 sectors excluding split $\pm 10^\circ$ is physical NOT_RUN. HOLD T1–T3: probe unselected; OD tolerance and insertion stop unqualified. Barrel flat-bottom bore 5.40 ± 0.05 preserves conservative ligament $3.345 \geq 3.32$, but does not qualify probe insertion; do not deepen or force probe to bottom. T4 insertion10/blind12 and T5 insertion4 are nominal candidates. Qualify retention, insulation and thermal response before assembly. SYS-04 die joint digital load-path PASS at1.50 N·m; physical receipt/leak/first thermal cycle NOT_RUN; shield clearance requires service verification. **단위:** mm · 제3각법 · NTS · source 2b085b5d04dda132909bd0e462a596893ad64912

16 FM-001 — cooling and strand path

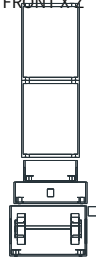
FM-001 cooling strand path

v0.8 · mm · 제3각법 · source: GGM integrated FreeCAD assembly · bodies 14

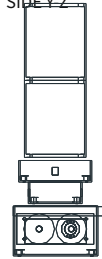
TOP X-Y



FRONT X-Z



SIDE Y-Z



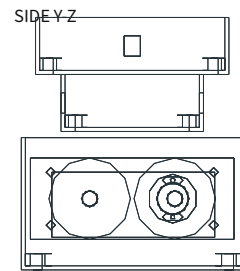
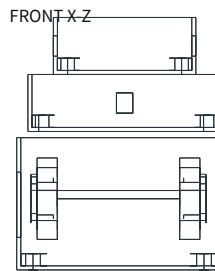
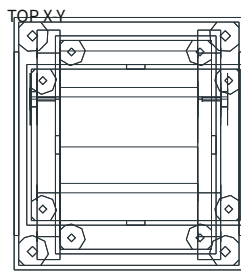
Overall X 114.00 · Y 125.00 · Z 328.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

재료/구성: 5052 duct / donor fans **핵심 치수/공차:** strand centreline offset ≤ 0.50 mm; hot-shield clearance ≥ 12 mm 단
위: mm · 제3각법 · NTS · source 2b085b5d04dda132909bd0e462a596893ad64912

17 FM-002 — gauge/puller

FM-002 gauge puller

v0.8 · mm · 제3각법 · source: GGM integrated FreeCAD assembly · bodies 11



Overall X 100.50 · Y 110.00 · Z 124.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

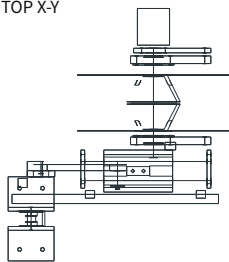
재료/구성: 6061 plates / silicone rollers / S45C eccentric bushes **핵심 치수/공차:** adjusted unloaded roller gap 1.60–1.90 mm over full rotation; bush index pair $\leq 0.5^\circ$; roller axes parallel $\leq 0.05/80$; gauge datum alignment ≤ 0.10 mm **단위:** mm · 제3각법 · NTS · source 2b085b5d04dda132909bd0e462a596893ad64912

18 SP-001 — spooler/traverse

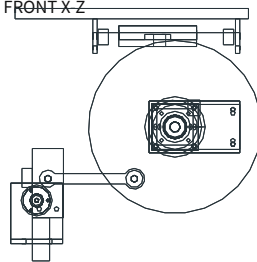
SP-001 spooler traverse

v0.8 · mm · 제3각법 · source: GGM integrated FreeCAD assembly · bodies 37

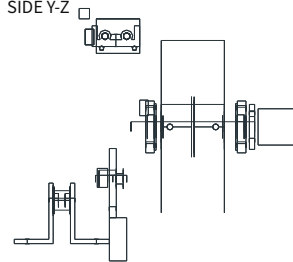
TOP X-Y



FRONT X-Z



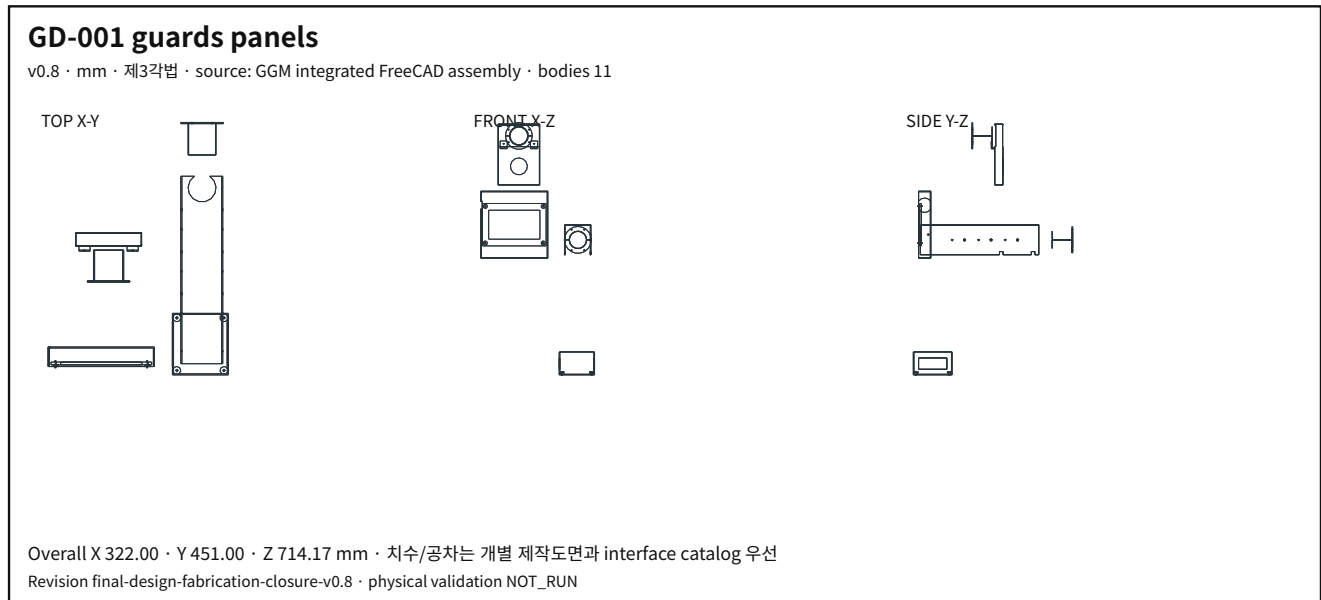
SIDE Y-Z



Overall X 290.00 · Y 330.00 · Z 292.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

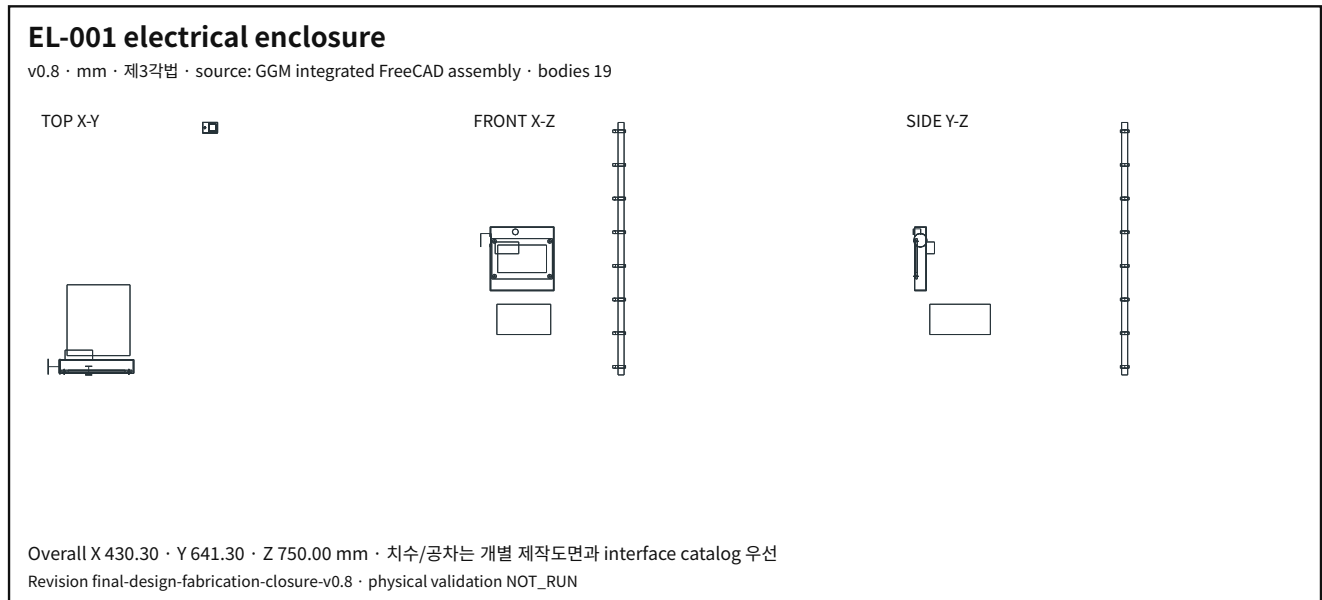
재료/구성: 6061 plates / stainless shafts **핵심 치수/공차:** spool shaft $\varnothing 12$ h6; traverse rod parallelism $\leq 0.10/160$ **단위:** mm · 제3각법 · NTS · source 2b085b5d04dda132909bd0e462a596893ad64912

19 GD-001 — guards and panels



재료/구성: polycarbonate and bonded metal panels **핵심 치수/공차:** hazard opening ≤ 6 mm; no reach path to moving/hot parts **단위:** mm · 제3각법 · NTS · source 2b085b5d04dda132909bd0e462a596893ad64912

20 EL-001 — electrical enclosure



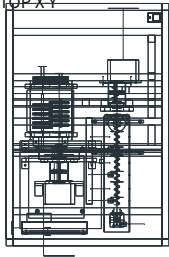
재료/구성: 2 mm 5052 enclosure **핵심 치수/공차:** PE bond target 0.10 ohm 이하; signal/power duct separation ≥ 18 mm
단위: mm · 제3각법 · NTS · source 2b085b5d04dda132909bd0e462a596893ad64912

21 SV-001 — service envelopes

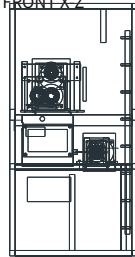
SV-001 service envelopes

v0.8 · mm · 제3각법 · source: GGM integrated FreeCAD assembly · bodies 184

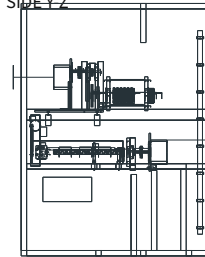
TOP X-Y



FRONT X-Z



SIDE Y-Z



Overall X 470.00 · Y 729.00 · Z 930.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

재료/구성: service-envelope reference geometry **핵심 치수/공차:** front/rear access ≥ 600 mm; hot-zone removal envelope kept clear **단위:** mm · 제3각법 · NTS · source 2b085b5d04dda132909bd0e462a596893ad64912